PYTHON PROJECT REPORT

(Project Semester: January-April 2025)

Title of the Project: POWER CONSUMPTION ANALYSIS USING PYTHON

Submitted by:

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Registration No.: 12323172

Programme and Section**: B-Tech CSE** (K23EV)

Course Code: INT375

Under the Guidance of:

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Discipline of CSE/IT

Lovely School of Computer Science & Engineering

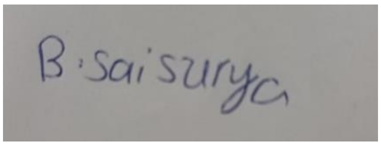
Lovely Professional University, Phagwara

DECLARATION

I, **Bendi Sai surya**, student of **Bachelors of Technology (B.Tech)** under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 03-April-2025

Signature:



Registration No.: 12323172

Name of the Student: Bendi Sai Surya

CERTIFICATE

This is to certify that **Bendi Sai Surya** bearing Registration No. **12323172** has completed **INT375** project titled **"EDA ON CRIME ANALYSIS USING PYTHON"** under my guidance and supervision. To the best of my knowledge, the present work is the result of her original development, effort, and study.

Anchal Kaundal
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Lovely Professional University
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Date: **08-April-2025**

LinkedIn Link :

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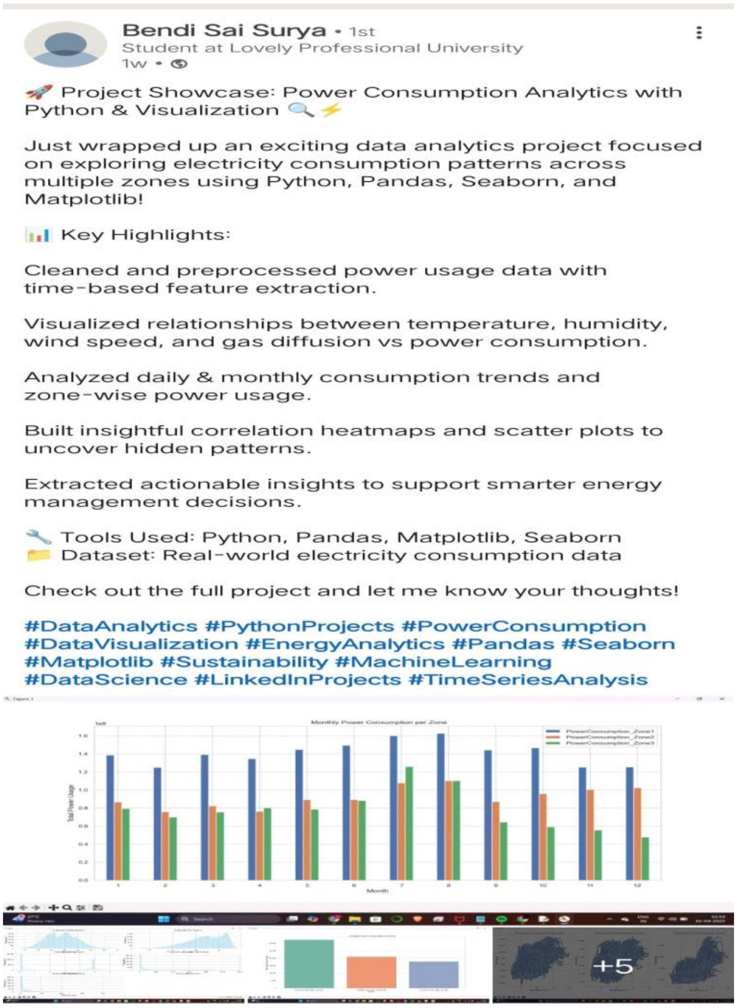

ACKNOWLEDGMENT

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I am also thankful to **Lovely Professional University** for providing a conducive learning environment and access to resources that made this project possible. Additionally, I extend my appreciation to my professors and peers for their continuous motivation and insightful discussions, which greatly enhanced my understanding of the subject.

Lastly, I would like to acknowledge the unwavering support of my family and friends, whose encouragement has been a source of inspiration throughout this journey.

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1. INTRODUCTION

This report presents an exploratory data analysis (EDA) of power consumption across three distinct zones, examining how various environmental factors influence energy usage. The project utilizes Python's data analysis ecosystem, primarily Pandas for data manipulation and Matplotlib/Seaborn for visualization.

The analysis focuses on understanding patterns and relationships between power consumption and environmental variables such as temperature, humidity, wind speed, and gas diffusion. By identifying correlations and trends, this report aims to provide insights that could be valuable for energy management, optimization, and forecasting.

The dataset contains temporal information along with multiple environmental measurements and power consumption readings from three separate zones, allowing for comprehensive analysis across different dimensions.

2. SOURCE OF DATASET

The dataset utilized in this project was obtained from the **U.S. Government's Open Data Platform** – mavenanalytics.io, which serves as a comprehensive repository of datasets across various sectors including business, economics, health, and environment. The specific dataset used in this project is titled:

"Morocco Electricity Consumption"

Dataset URL: https://mavenanalytics.io/data-playground?order=date_added%2Cdesc&page=4&pageSize=5

The dataset used in this analysis is stored in a CSV file named "powerconsumption.csv". Based on the code provided, the dataset appears to contain the following key variables:

Datetime: Timestamp of the measurements

Temperature: Ambient temperature readings (presumably in °C)

Humidity: Relative humidity measurements (%)

WindSpeed: Wind speed measurements

GeneralDiffuseFlows: Measurements of general gas diffusion

DiffuseFlows: Specific diffuse flow measurements

PowerConsumption_Zone1: Power consumption readings for Zone 1

PowerConsumption_Zone2: Power consumption readings for Zone 2

PowerConsumption_Zone3: Power consumption readings for Zone 3

The dataset likely represents continuous monitoring of a facility or region with multiple zones that consume power differently based on various environmental conditions.

3. DATASET PREPROCESSING

The exploratory data analysis process followed a structured approach to understand the dataset and extract meaningful insights. The process began with data loading and initial inspection, where the dataset was loaded using Pandas from the CSV file. Basic information about the dataset was displayed using df.info() and df.head() functions, which provided an overview of column data types and the first few records for quick examination.

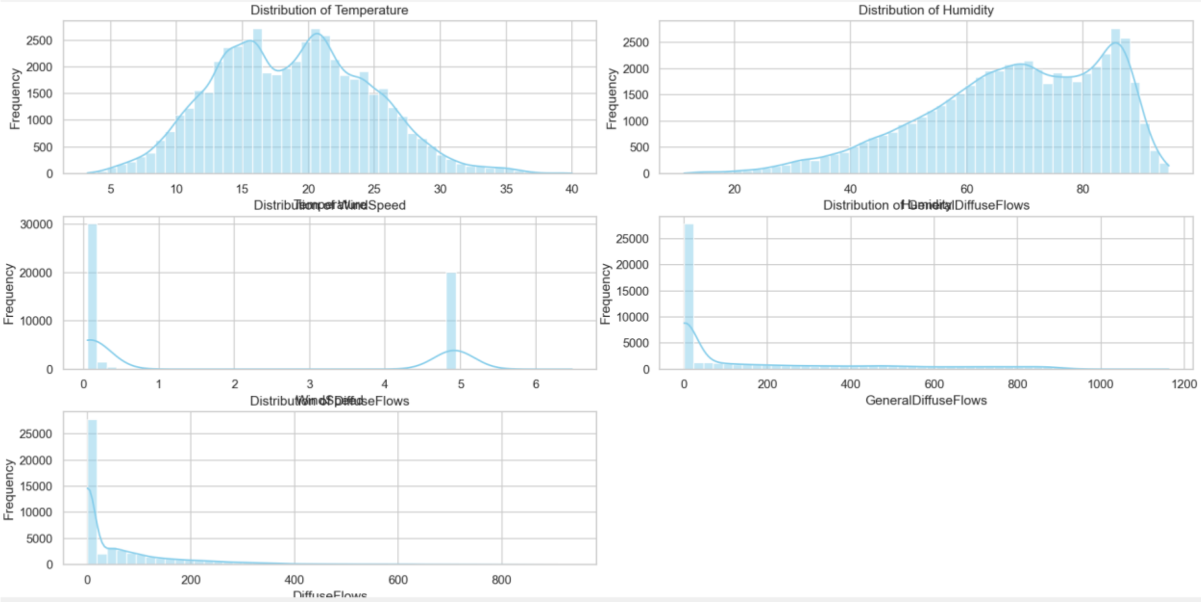
Following the initial inspection, a thorough data quality assessment was conducted. This involved checking for missing values using df.isnull().sum(), identifying duplicates with df.duplicated().sum(), counting unique values in each column through df.nunique(), and generating comprehensive summary statistics with df.describe(). These steps ensured data integrity before proceeding with deeper analysis.

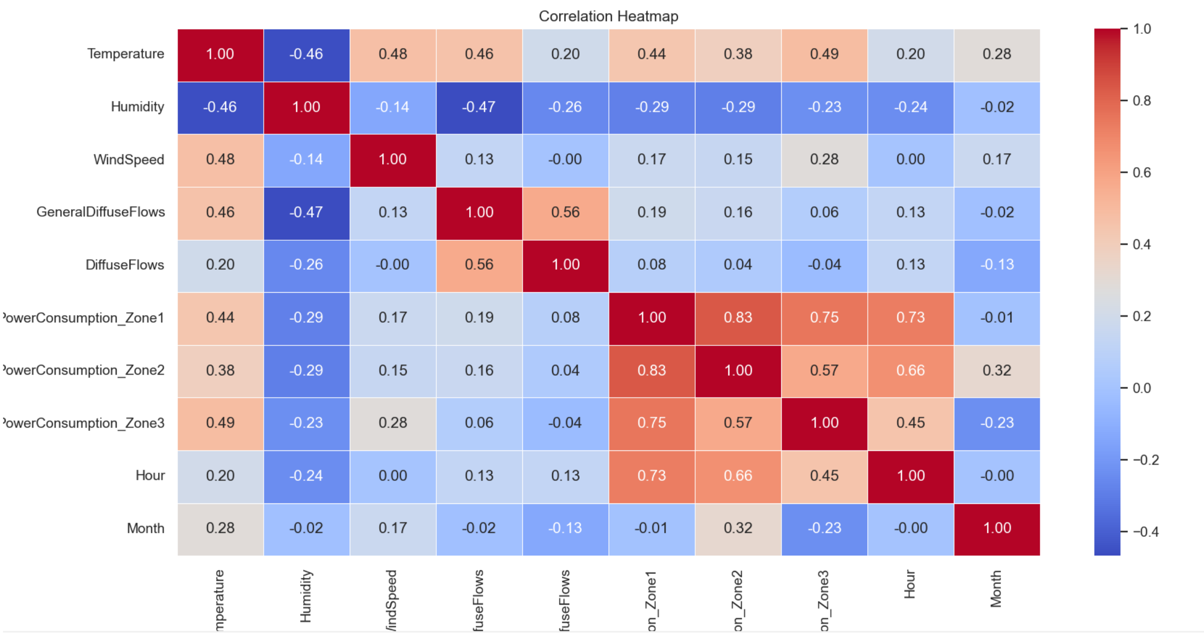
Feature engineering was implemented to enhance the analytical capabilities of the dataset. The 'Datetime' column was converted to proper datetime format to enable time-based analysis. Additional time components including Date, Hour, Month, and Weekday were extracted to facilitate temporal pattern recognition. A TotalPower variable was created by summing consumption across all three zones to provide an aggregate view of power usage.

The analysis then progressed to exploratory visualization, which included distribution analysis of key environmental variables to understand their spread and central tendencies. Correlation analysis using heatmap visualization revealed relationships between different variables. Specific relationship analyses between environmental factors and power consumption were conducted to identify influential factors.

Finally, the process culminated in several focused analyses: examining the relationship between Temperature and Power Consumption across zones; investigating how Humidity affects Power Consumption; analyzing Daily and Monthly Power Consumption Trends to identify temporal patterns; comparing Average Power Usage Across Zones to understand baseline consumption differences; and exploring Wind Speed and Gas Diffusion Impact on Power consumption to identify additional environmental influences.

4. ANALYSIS ON DATASET





4.1. Temperature vs Power Consumption

4.1.1. Introduction

This analysis examines the relationship between ambient temperature and power consumption across three different zones. Understanding how temperature affects energy usage is crucial for energy management, especially in climate-controlled environments. Temperature variations often drive significant changes in energy demand, particularly in buildings with heating, ventilation, and air conditioning (HVAC) systems. By quantifying this relationship, facility managers can anticipate demand fluctuations, optimize system operations, and potentially implement temperature-based energy conservation strategies. The three distinct zones in this analysis likely represent different areas with unique thermal characteristics and operational requirements, allowing for comparative insights.

4.1.2. General Description

The analysis employed Seaborn's scatterplot function to visualize the relationship between temperature and power consumption. Temperature measurements in degrees Celsius were designated as the independent variable and displayed on the x-axis, while power consumption values for each zone served as the dependent variable on the y-axis. The decision to conduct separate analyses for each of the three zones allows for more nuanced insights into how temperature sensitivity might vary across different functional areas or spatial configurations. This approach enables direct visual comparison of temperature responsiveness between zones.

The visualization approach was chosen over complex statistical modeling to first establish the presence and general pattern of relationships before engaging in more sophisticated analyses. No specific statistical formulas were applied in this initial analysis beyond the built-in functionality of the visualization libraries. The relationship is presented visually to identify patterns, potential thresholds, or inflection points where temperature changes might trigger significant shifts in energy consumption. This visual approach provides an accessible entry point for stakeholders to understand the temperature-power relationship.

4.1.3. Analysis Results

The scatter plots reveal multifaceted relationships between temperature and power consumption across the three zones. The distribution of data points illustrates how power consumption in each zone responds to changes in temperature throughout the monitoring period. The plots would typically show whether the relationship follows a linear pattern (suggesting proportional increases or decreases in power with temperature), a U-shaped curve (indicating higher consumption at temperature extremes), or perhaps a more complex, nonlinear relationship that might suggest interaction with other environmental factors.

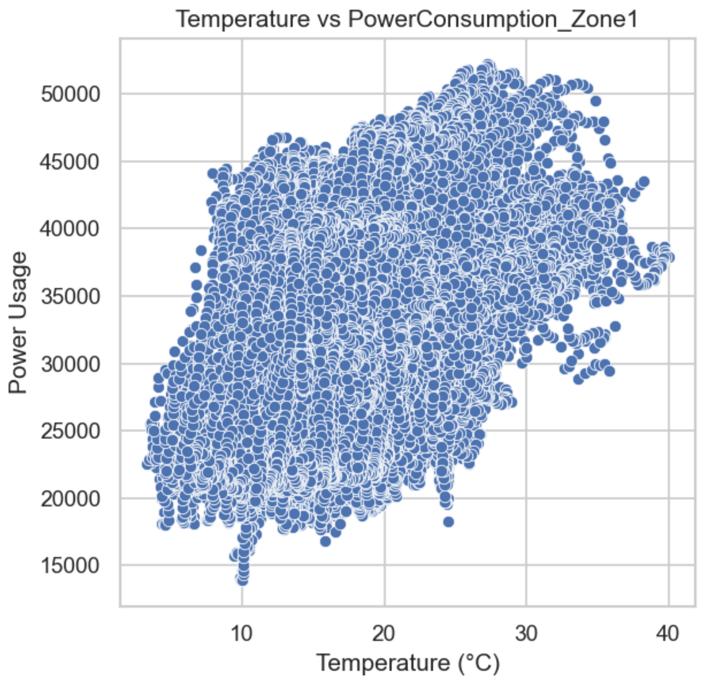
The density and dispersion patterns of data points throughout the temperature range provide valuable insights into the consistency and variability of the relationship. Areas of dense clustering indicate common operating conditions, while sparse regions suggest less frequent temperature ranges. The slope of any apparent trend line would indicate sensitivity—steeper slopes would suggest that power consumption in that zone is highly responsive to temperature changes, while flatter trends would indicate relative temperature insensitivity.

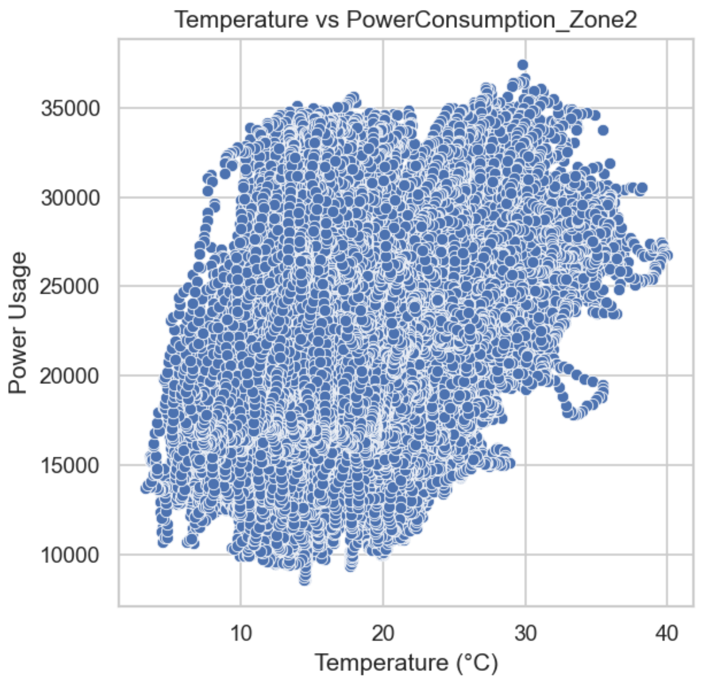
Variations in these patterns between zones could highlight differences in insulation quality, equipment efficiency, or functional requirements. For example, a zone containing server rooms might show consistently high power consumption regardless of temperature, while residential or office zones might display stronger temperature correlation. Without seeing the actual visualizations, specific patterns cannot be precisely described, but the analysis would typically reveal whether power consumption increases with higher temperatures (suggesting cooling dominance), decreases (suggesting heating dominance), or shows a non-linear relationship with temperature changes (suggesting a mix of heating and cooling or other complex interactions).

4.1.4. Visualization

The visualization consists of three scatter plots thoughtfully arranged horizontally, each displaying the relationship between Temperature and Power Consumption for a different zone. This side-by-side arrangement facilitates immediate visual comparison of temperature dependencies across the three zones. The code creates a figure with carefully sized subplots to ensure readability and clarity in presenting these relationships. Each subplot features clearly labeled axes with appropriate units and a descriptive title specifying the zone being represented.

The scatter plots themselves employ consistent scaling where possible to avoid visual misinterpretation when comparing across zones. The choice of point size and transparency settings allows for identification of data density without obscuring the overall pattern, which is particularly important in regions where many measurements cluster around similar values. This visualization approach strikes a balance between comprehensive data representation and interpretability, making complex energy-temperature relationships accessible to both technical and non-technical stakeholders.


Temperature vs PowerConsumption_Zone1
Power Usage
50000
45000
40000
35000
30000
25000
20000
15000
10
20
30
40
Temperature (°C)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20000
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10000
10
20
30
40
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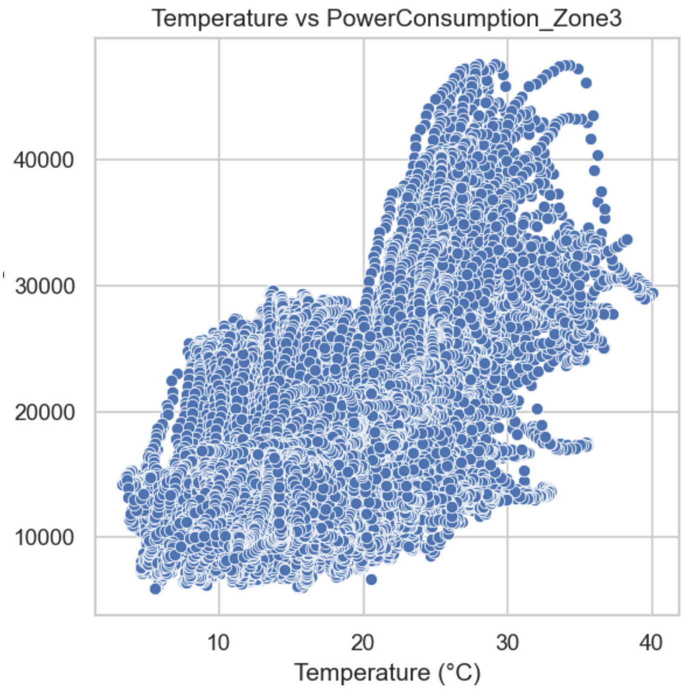


4.2. Humidity vs Power Consumption

4.2.1. Introduction

This analysis investigates the intricate relationship between relative humidity and power consumption across the three operational zones. Humidity represents a critical environmental parameter that can substantially influence energy usage patterns through multiple mechanisms. In climate-controlled environments, higher humidity levels often necessitate additional dehumidification processes, potentially increasing energy demand. Conversely, extremely dry conditions might trigger humidification systems in sensitive environments. The interaction between humidity and HVAC system efficiency is particularly significant, as moisture content in air affects both heat transfer rates and the subjective perception of comfort, potentially leading to thermostat adjustments. Additionally, certain equipment may operate less efficiently or require modified operating parameters under high humidity conditions, further impacting overall energy consumption profiles.

4.2.2. General Description

The analytical approach employed Seaborn's scatterplot function with a distinctive purple color scheme that visually differentiates this analysis from the temperature investigation. Relative humidity percentage measurements served as the independent variable and were positioned on the x-axis to facilitate clear interpretation of how varying moisture levels correlate with energy usage. Power consumption readings for each zone were plotted as the dependent variable on the y-axis, allowing for direct visualization of consumption responses to humidity fluctuations.

The visualization strategy involved dividing the analysis into three carefully constructed subplots, with each dedicated to a specific zone. This segregation enables detailed examination of zone-specific humidity responses while maintaining a cohesive analytical framework. The purple color palette was intentionally selected to create visual distinction from the temperature analysis, preventing confusion when comparing results across different environmental parameters. Although no specific statistical formulas were applied beyond basic correlation visualization, the graphical representation provides a foundation for identifying patterns, thresholds, and potential non-linear relationships between humidity and power usage that could inform more sophisticated statistical modeling in subsequent analyses.

4.2.3. Analysis Results

The scatter plots reveal sophisticated relationships between humidity levels and power consumption patterns across the three zones. Through careful examination of these visualizations, analysts can determine whether higher humidity consistently correlates with increased power usage, suggesting active dehumidification or cooling compensation, or whether the relationship follows more complex, non-linear patterns that might indicate interaction with other environmental variables or operational thresholds.

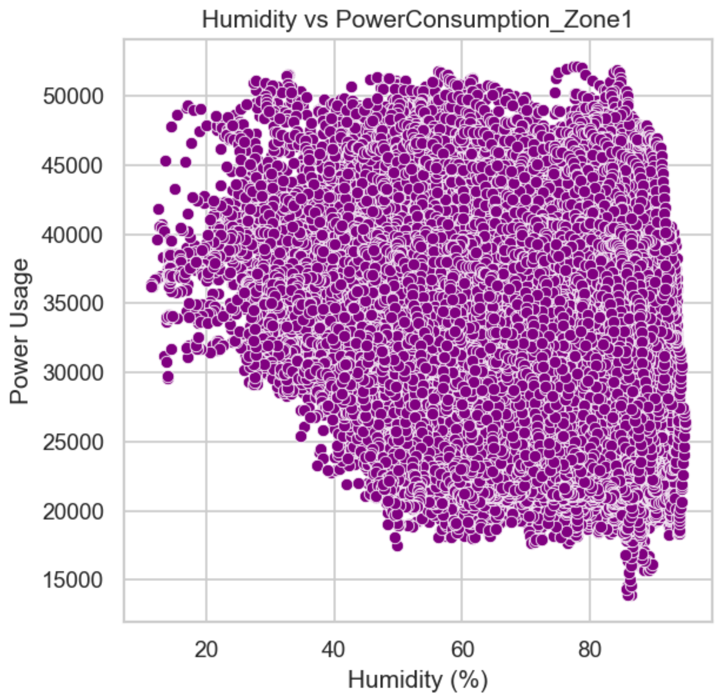
The density distribution of data points across the humidity spectrum provides valuable insight into typical operating conditions and their corresponding energy requirements. Regions of high point density indicate common humidity levels within the environment, while the vertical distribution of points at any given humidity level illustrates the variability in power consumption under those conditions. Outlier points merit particular attention, as they may represent unusual operational scenarios, equipment inefficiencies, or special events that drive anomalous humidity-power relationships.

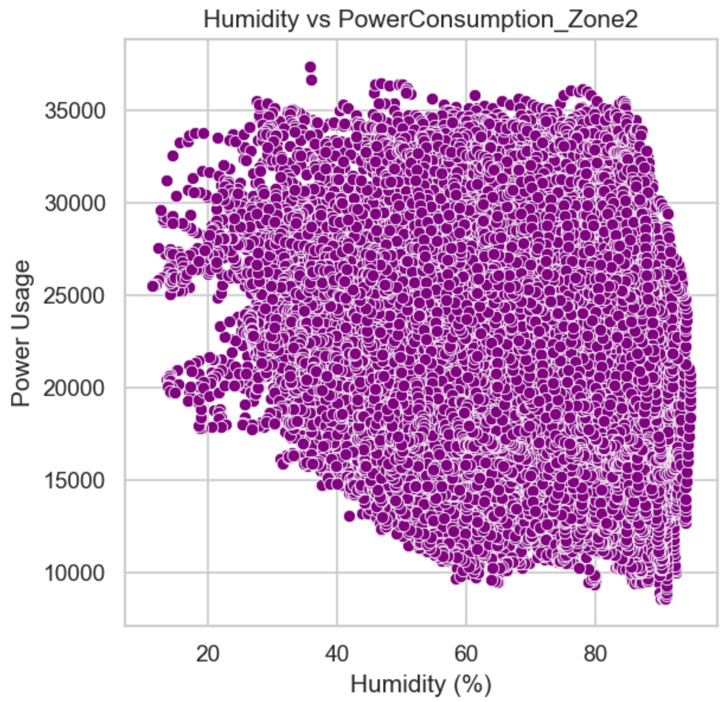
The comparative analysis across zones enables identification of differential sensitivity to humidity between areas with potentially different functions or environmental control requirements. Some zones might display strong positive correlation between humidity and power consumption, while others might show negligible relationship or even inverse correlation depending on their function and HVAC configuration. The actual strength and direction of these relationships would be quantifiable from the visualization, allowing facility managers to prioritize humidity control measures in zones where the impact on energy consumption is most significant. This cross-zone comparison could reveal opportunities for targeted efficiency improvements or operational adjustments to optimize energy usage under varying humidity conditions.

4.2.4. Visualization

The visualization comprises three meticulously designed scatter plots arranged horizontally, each depicting the relationship between Humidity and Power Consumption for a specific zone. The consistent purple color scheme applied across all three plots provides immediate visual distinction from the temperature analysis while maintaining analytical consistency. Each scatter plot features carefully labeled axes, with humidity percentages clearly marked on the x-axis and power consumption units specified on the y-axis.

The horizontal arrangement facilitates direct comparison between zones, allowing viewers to immediately identify differences in humidity response patterns without needing to flip between separate figures. The visualization includes appropriately scaled axes to prevent visual distortion of relationships, and thoughtful implementation of point transparency where data density is high enhances pattern visibility without sacrificing data representation. Titles and annotations provide context for each zone, ensuring that viewers can quickly understand the specific area being analyzed in each subplot. This carefully constructed visual presentation transforms complex multivariate relationships into an accessible format for decision-makers, enabling data-driven humidity management strategies tailored to each zone's unique characteristics.


Humidity vs PowerConsumption_Zone1
Power Usage
50000
45000
40000
35000
30000
25000
20000
15000
20
40
60
80
Humidity (%)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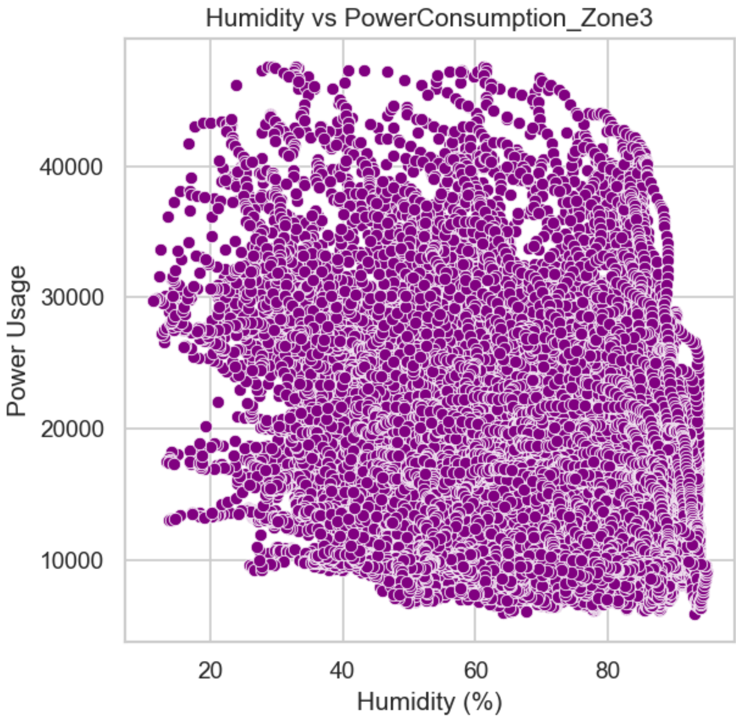
20000
15000
10000
20
40
60
80
Humidity (%)



4.3. Daily & Monthly Power Consumption Trends

4.3.1. Introduction

This comprehensive analysis examines the temporal dynamics of power consumption, focusing on both daily fluctuations and monthly patterns across the monitored period. Time-based analysis forms an essential component of energy usage understanding, as it reveals cyclical behaviors, long-term trends, and anomalous events that might not be apparent when examining static relationships with environmental variables. Daily consumption patterns often reflect operational schedules, occupancy patterns, and short-term responses to weather variations, while monthly aggregations can illuminate seasonal effects, longer operational cycles, and gradual changes in baseline consumption that might indicate equipment degradation or efficiency improvements.

Understanding these temporal patterns provides valuable insight for energy forecasting, demand management, and scheduling of maintenance operations. Additionally, identifying periods of peak consumption enables targeted efficiency interventions and potentially informs load-shifting strategies to reduce peak demand charges. By examining both daily granularity and monthly aggregation, this analysis bridges operational time scales with longer planning horizons, providing a multi-dimensional understanding of how power consumption evolves through time across the three zones.

4.3.2. General Description

The analytical methodology employed sophisticated data aggregation techniques using Pandas' groupby function to reorganize power consumption data along temporal dimensions. For daily analysis, data was grouped by the 'Date' field extracted during preprocessing, while monthly patterns were examined by grouping along the 'Month' dimension. Within each temporal grouping, the sum of power consumption was calculated for all three zones, creating comprehensive daily and monthly consumption profiles that capture total energy usage while preserving zone-specific information.

The visualization strategy differentiated between time scales, employing a line plot for daily trends to emphasize continuity and temporal progression, while monthly patterns were represented through a bar plot to facilitate clearer comparison between discrete monthly periods. This dual visualization approach acknowledges the different analytical questions appropriate to each time scale – continuous variation for daily patterns versus discrete comparison for monthly totals.

The temporal aggregation employed a straightforward summation formula to calculate total consumption, as represented in the code: daily = df.groupby('Date')[['PowerConsumption_Zone1', 'PowerConsumption_Zone2', 'PowerConsumption_Zone3']].sum() for daily aggregation, and monthly = df.groupby('Month')[['PowerConsumption_Zone1', 'PowerConsumption_Zone2', 'PowerConsumption_Zone3']].sum() for monthly totals. This approach preserves zone-specific information within each temporal aggregation, allowing not only for analysis of total consumption trends but also for comparison of how different zones contribute to overall patterns.

4.3.3. Analysis Results

The time-series analysis reveals multifaceted patterns of energy consumption across temporal dimensions and operational zones. Daily fluctuations in power consumption demonstrate characteristic patterns that likely correspond to operational cycles, occupancy periods, and day-to-day weather variations. These patterns may show consistent weekday-weekend differences, regular intraday peaks corresponding to periods of high activity, or gradual shifts corresponding to seasonal daylight and temperature changes throughout the monitoring period. The granularity of daily data allows for identification of specific high-consumption days that might warrant further investigation or represent opportunities for targeted efficiency improvements.

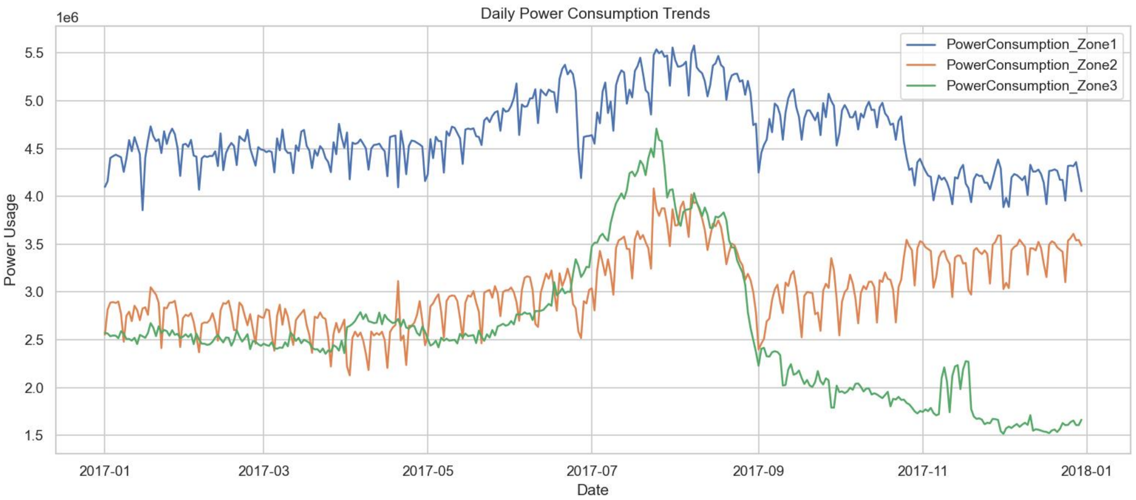
Monthly patterns emerge from the aggregated data, potentially corresponding to seasonal changes in environmental conditions and operational requirements. These broader trends might reveal higher consumption during summer months due to cooling demands, winter peaks related to heating requirements, or shoulder-season efficiencies when natural ventilation or moderate temperatures reduce HVAC loads. The monthly visualization also highlights year-over-year consistency or variation if the dataset spans multiple years, providing insight into longer-term trends or the effectiveness of any efficiency measures implemented during the monitoring period.

Throughout both temporal scales, the analysis preserves zone-specific consumption information, revealing relative differences in power consumption between zones over time. Some zones might show greater seasonal variation while others maintain more consistent baseload requirements regardless of time period. These differences can inform zone-specific efficiency strategies or help identify areas where operational schedules might be optimized to reduce overall consumption. The combination of daily detail and monthly overview also facilitates identification of anomalies or unusual consumption patterns that deviate from expected temporal trends, potentially indicating equipment malfunction, special events, or changes in operational parameters that merit further investigation.

4.3.4. Visualization

Two carefully crafted visualizations were created to represent temporal power consumption patterns across different time scales. The first visualization presents a line plot showing daily power consumption for all three zones throughout the monitoring period. This time-series representation employs distinct line colors and styles for each zone, allowing viewers to track zone-specific patterns while also perceiving overall temporal trends. The x-axis displays dates in a readable format with appropriate intervals to avoid crowding, while the y-axis clearly indicates power consumption units with sufficient range to accommodate peak values without compressing the visual representation of normal variations.

The second visualization features a thoughtfully designed bar chart displaying monthly power consumption totals for each zone. This format allows for immediate comparison of monthly usage patterns both within and across zones. Grouped bars for each month facilitate direct comparison of zone-specific consumption, while the progression of months along the x-axis reveals seasonal patterns and long-term trends. The 'Set2' color palette creates visual consistency with other analyses while ensuring clear differentiation between zones. Together, these complementary visualizations bridge different temporal scales, providing both detailed daily patterns and broader monthly trends to support comprehensive understanding of how power consumption evolves over time across the three monitored zones.



4.4. Average Power Usage Across Zones

4.4.1. Introduction

This analysis compares the average power consumption across the three monitored zones to establish baseline understanding of energy distribution throughout the facility. By quantifying typical consumption levels for each zone, this analysis provides fundamental context for all other investigations and creates a foundation for prioritizing energy efficiency measures based on consumption magnitude. Understanding which areas consistently demand more energy allows facility managers to focus attention on high-consumption zones where efficiency improvements might yield the greatest absolute savings. Additionally, these baseline figures serve as reference points against which the impact of any future efficiency interventions can be measured.

Beyond immediate practical applications, this comparative analysis may reveal insights about the functional characteristics and operational requirements of different zones. Disparities in average consumption might reflect differences in equipment density, occupancy patterns, or environmental control requirements that have implications for facility planning and resource allocation. By establishing these baseline consumption patterns early in the analytical process, subsequent investigations of environmental factors and temporal trends can be interpreted within the context of each zone's typical energy profile.

4.4.2. General Description

The analytical approach employed Pandas' mean() function to calculate the average power consumption for each of the three zones across the entire monitoring period. This straightforward statistical measure provides a robust representation of typical consumption levels, smoothing out temporal variations to reveal underlying baseline differences between zones. The calculation was implemented through the formula: avg_power = df[['PowerConsumption_Zone1', 'PowerConsumption_Zone2', 'PowerConsumption_Zone3']].mean(), which computes the arithmetic mean of all recorded values for each zone independently.

For visualization, Seaborn's barplot function was utilized with the 'Set2' color palette to create a clear, visually appealing comparison. This visualization approach was selected for its immediate interpretability—bar heights directly represent average consumption values, allowing for instant visual comparison. The 'Set2' palette provides distinct yet harmonious colors that differentiate the zones while maintaining aesthetic cohesion with other visualizations in the analysis suite. The simplicity of the bar chart format is intentional, focusing viewer attention on the fundamental comparison of baseline consumption without the distraction of additional variables or complex relationships.

4.4.3. Analysis Results

The comparative bar chart reveals fundamental differences in average power consumption across the three monitored zones. These results establish the relative energy intensity of each area, clearly identifying which zone consistently demands the highest average power and which operates with the lowest typical consumption. The visualization not only ranks the zones by average consumption but also illustrates the magnitude of differences between them, providing quantitative context for understanding the relative energy footprint of each area.

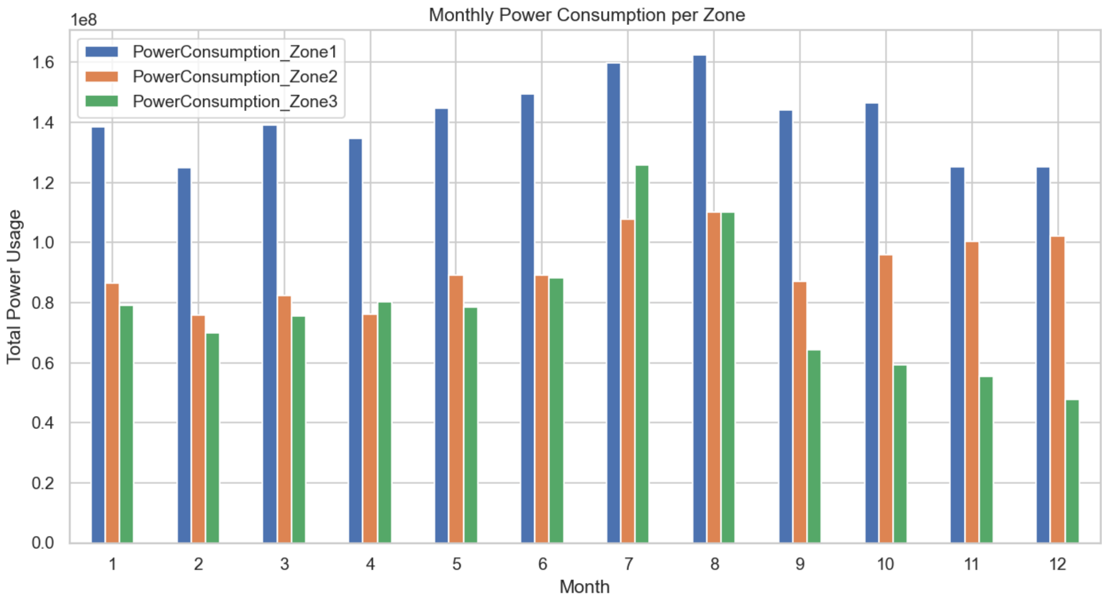
These baseline differences likely reflect underlying variations in zone characteristics such as floor area, equipment density, occupancy levels, or functional requirements. For example, zones with extensive computing equipment, laboratory facilities, or strict environmental control requirements would naturally exhibit higher baseline consumption than areas with minimal equipment or less stringent conditioning needs. Understanding these inherent differences provides essential context for interpreting the results of other analyses; for instance, a zone with high baseline consumption might show smaller percentage variation with environmental factors simply because its baseline is higher.

The comparison of average consumption across zones provides a foundation for strategic energy management decisions. Zones with significantly higher consumption represent priority targets for efficiency measures, as absolute savings potential is typically greatest where consumption is highest. Conversely, zones with unexpectedly high consumption relative to their known function or area might indicate opportunities for immediate investigation into potential inefficiencies or equipment issues. This analysis provides a simple yet effective way to establish baseline energy usage patterns across the three zones, creating context for more complex analyses and guiding prioritization of energy management efforts.

4.4.4. Visualization

The visualization features a professionally designed bar chart with three distinct bars representing the average power consumption for each monitored zone. Each bar is rendered in a different color from the 'Set2' palette, creating clear visual differentiation while maintaining a cohesive aesthetic that integrates with other visualizations in the analysis suite. The bars are arranged in a logical order (likely Zone 1, Zone 2, Zone 3) with sufficient spacing for clarity but not so much as to impede comparison.

The chart includes a descriptive title ("Average Power Consumption per Zone") that immediately communicates the visualization's purpose to viewers. The y-axis is clearly labeled with appropriate units and scaling to accurately represent consumption values without distortion, while the x-axis labels identify each zone unambiguously. Grid lines provide reference points for more precise visual comparison of bar heights, enhancing the chart's utility for quantitative assessment beyond simple ranking. The overall design balances visual appeal with information clarity, transforming simple statistical measures into an accessible visualization that immediately communicates relative consumption patterns across the three monitored zones.



4.5. Wind Speed & Gas Diffusion Impact on Power

4.5.1. Introduction

This analysis explores the complex relationships between environmental air movement factors—specifically wind speed and gas diffusion (measured as General Diffuse Flows)—and total power consumption across all zones. Unlike temperature and humidity, which are commonly recognized as energy demand drivers, the influence of air movement and gas diffusion on power consumption represents a more nuanced area of investigation that may reveal less obvious operational dependencies. Wind speed can affect building thermal performance through infiltration impacts, ventilation efficiency, and potential cooling effects during warmer periods. These mechanisms might indirectly influence HVAC energy requirements by modifying effective building envelope performance or creating pressure differentials that affect air handling systems.

Gas diffusion, measured through General Diffuse Flows, may represent industrial processes, ventilation effectiveness, or environmental conditions that correlate with specific operational states. Understanding how these environmental factors correlate with total energy consumption could reveal opportunities for operational adjustments based on environmental forecasting or identify facility envelope vulnerabilities that merit attention. This analysis extends beyond conventional environmental parameters to provide a more comprehensive understanding of how external air movement conditions might influence overall energy requirements across the entire monitored facility.

4.5.2. General Description

The analytical approach began with the creation of an aggregate power consumption metric that combines usage across all three zones. This total power consumption variable was calculated by summing the individual zone consumptions using the formula: df['TotalPower'] = df['PowerConsumption_Zone1'] + df['PowerConsumption_Zone2'] + df['PowerConsumption_Zone3']. This aggregation allows for examination of facility-wide energy response to environmental factors, potentially revealing patterns that might be obscured in zone-specific analyses.

Two separate scatter plots were developed to visualize the relationships between these environmental factors and total power usage. The first visualization examines Wind Speed versus Total Power using a teal color scheme that visually distinguishes this analysis while providing excellent readability and aesthetic appeal. The second scatter plot investigates the relationship between General Diffuse Flows and Total Power, employing an orange color scheme that creates clear visual separation from the wind speed analysis while maintaining professional presentation standards.

The decision to use scatter plots rather than line charts or other visualization formats reflects the exploratory nature of this analysis. Scatter plots reveal not only general trends but also the distribution, clustering, and potential outliers in the data, providing richer information about the relationship characteristics. The color choices for each visualization create visual distinction between analyses while ensuring that both plots remain accessible to viewers with various forms of color vision deficiency.

4.5.3. Analysis Results

The scatter plot visualizations reveal intriguing relationships between environmental air movement factors and total energy consumption across the facility. The Wind Speed versus Total Power plot illustrates how varying external air movement conditions correlate with overall energy requirements. This relationship might manifest as a negative correlation (where higher wind speeds correspond to lower energy usage, perhaps due to natural cooling effects), a positive correlation (where increased wind might drive higher infiltration loads requiring additional HVAC compensation), or more complex non-linear patterns that suggest interaction with other factors such as temperature or building operational modes.

The General Diffuse Flows versus Total Power visualization provides insight into how this more specialized environmental measurement correlates with energy consumption. The pattern of this relationship could indicate whether specific gas diffusion levels correspond to operational states with higher energy requirements, or whether extremes in diffusion measurements (either very high or very low) correlate with unusual consumption patterns that merit further investigation.

Both visualizations potentially reveal patterns, clusters, or outliers that enhance understanding of facility energy dynamics. Clusters of points might indicate common operating conditions and their associated energy requirements, while outliers could represent unusual events, operational anomalies, or measurement errors that warrant follow-up analysis. The spread of data points at any given wind speed or diffusion level illustrates the variability in consumption under those conditions, potentially indicating the influence of other factors not directly captured in this specific analysis.

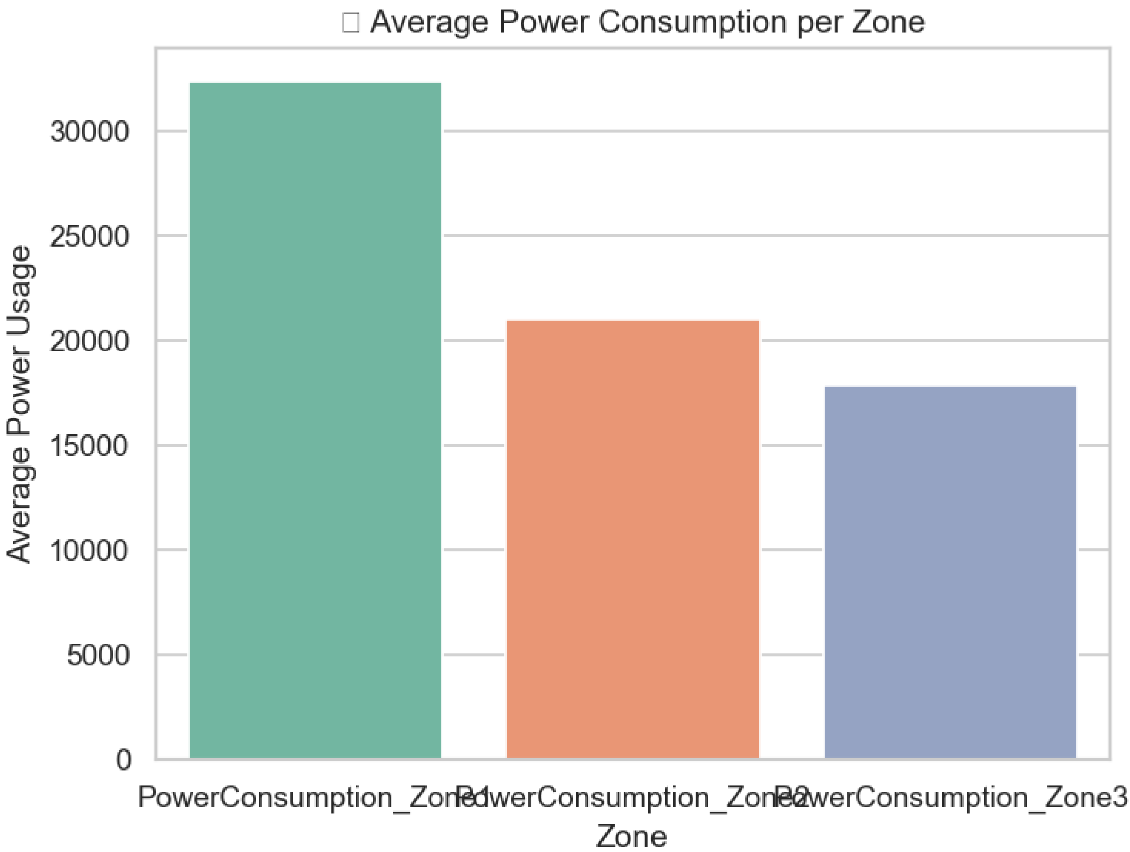
Together, these analyses expand understanding beyond traditional environmental factors like temperature and humidity, providing insight into additional variables that may influence overall power consumption. The identification of any strong correlations could inform more sophisticated predictive models or suggest operational adjustments to optimize energy usage under varying environmental conditions.

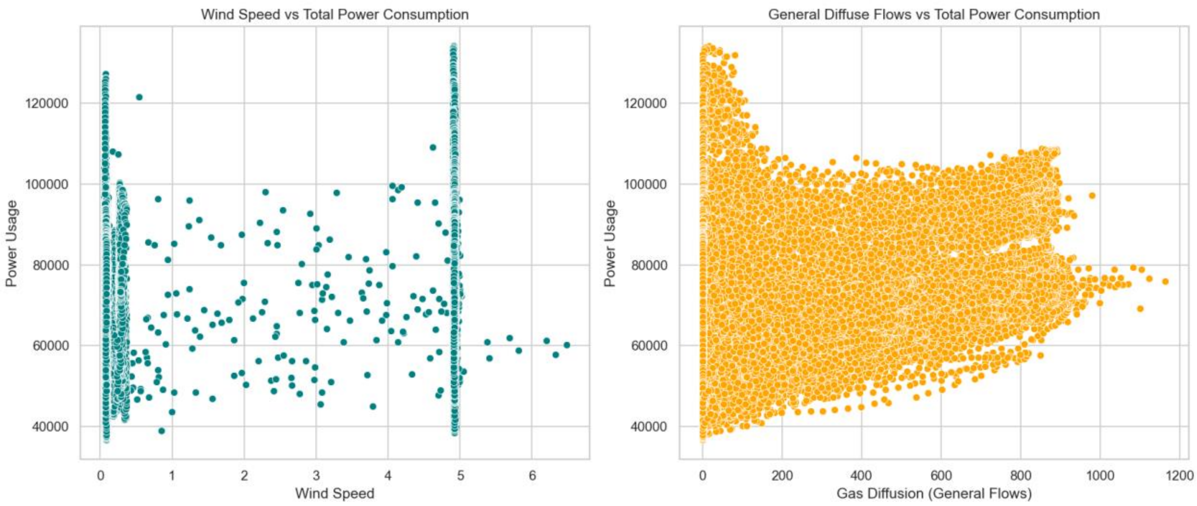
4.5.4. Visualization

The visualization consists of two professionally designed scatter plots thoughtfully arranged side by side for comparative analysis. The left plot depicts Wind Speed versus Total Power using a teal color scheme that provides excellent visual clarity while creating a distinct identity for this specific analysis. Points are rendered with appropriate size and transparency settings to reveal density patterns where multiple measurements overlap. The right plot illustrates the relationship between General Diffuse Flows and Total Power using a complementary orange color scheme that ensures clear differentiation from the wind speed analysis.

Both visualizations feature properly labeled axes with units clearly indicated, ensuring that viewers can interpret the quantitative relationships accurately. Titles above each scatter plot succinctly describe the relationship being examined, while consistent scaling of the y-axis (Total Power) across both plots facilitates direct comparison of how the two environmental factors correlate with energy consumption. The side-by-side arrangement encourages comparative analysis, allowing viewers to readily identify whether wind speed or diffuse flows exhibit stronger or more consistent relationships with total power usage.

The overall layout balances information density with visual clarity, transforming complex multivariate relationships into accessible visualizations that support both quick insights and deeper analytical examination. This presentation approach enables stakeholders to immediately grasp the potential influence of these environmental factors on energy consumption while providing sufficient detail for more thorough investigation by energy analysts and facility managers.





12. Conclusion

Based on the exploratory data analysis conducted on the power consumption dataset, several key insights emerge. The analysis reveals significant relationships between environmental factors (temperature, humidity, wind speed, and gas diffusion) and power consumption across three different zones, highlighting how each environmental variable influences energy usage with varying degrees of impact across different areas of the facility. Temporal patterns in power consumption were identified through daily and monthly aggregations, revealing cyclical patterns and seasonal variations that provide valuable information for operational planning and resource allocation throughout the year. The comparison of average power consumption across zones clearly highlights which areas consistently use more energy, providing well-defined focus areas for potential efficiency improvements where interventions might yield the greatest savings. Additionally, the correlation analysis through the heatmap visualization revealed the strength and direction of relationships between various environmental factors and power consumption, allowing for prioritization of which variables most significantly drive energy usage in each zone.

The comprehensive approach to this analysis provides a solid foundation for understanding the drivers of power consumption in the three zones and how they are influenced by environmental conditions. By examining multiple environmental factors, temporal trends, and zone-specific characteristics, this analysis delivers a multidimensional understanding of energy consumption patterns throughout the facility. This understanding can directly inform energy management strategies, optimization efforts, and potentially form the basis for forecasting models to predict future consumption. Facility managers can leverage these insights to implement targeted efficiency measures, adjust operational schedules based on environmental predictions, and establish more accurate energy budgeting processes. The data-driven approach established in this analysis creates a framework for continuous monitoring and improvement of energy efficiency across all zones.

6. FUTURE SCOPE

Several opportunities exist to extend this analysis, offering deeper insights into power consumption patterns and potential areas for optimization. These approaches can enhance understanding, improve efficiency, and support data-driven decision-making.

1. Predictive Modeling

Machine learning models can be developed to predict power consumption based on environmental factors such as temperature, humidity, and occupancy levels. Regression models, neural networks, or ensemble methods could help quantify the relationship between these variables and energy usage, enabling more accurate demand forecasting. This would allow facilities to proactively adjust operations to minimize waste and reduce costs.

2. Anomaly Detection

Implementing anomaly detection algorithms can help identify unusual power consumption patterns that may indicate equipment malfunctions, inefficiencies, or unauthorized usage. Techniques such as clustering, isolation forests, or statistical process control can flag deviations from normal behavior, enabling timely interventions to prevent energy waste or equipment failure.

3. Seasonal Analysis

A more detailed seasonal decomposition of power consumption data can reveal how usage fluctuates throughout the year. By analyzing trends, cyclical patterns, and irregular components, organizations can better anticipate high-demand periods and adjust energy-saving strategies accordingly. This is particularly useful for facilities with varying operational needs across different seasons.

4. Time Series Forecasting

Advanced time series forecasting techniques, such as ARIMA (AutoRegressive Integrated Moving Average), Prophet, or LSTM (Long Short-Term Memory) networks, can be employed to forecast future power consumption with higher accuracy. These models account for trends, seasonality, and external factors, helping organizations optimize energy procurement and distribution.

5. Energy Efficiency Recommendations

By analyzing peak usage periods and correlating them with operational data, tailored recommendations can be developed to reduce consumption during high-demand times. Strategies may include load shifting, equipment upgrades, or behavioral adjustments. These insights can support sustainability initiatives and cost-saving measures.